

Electric Circuits



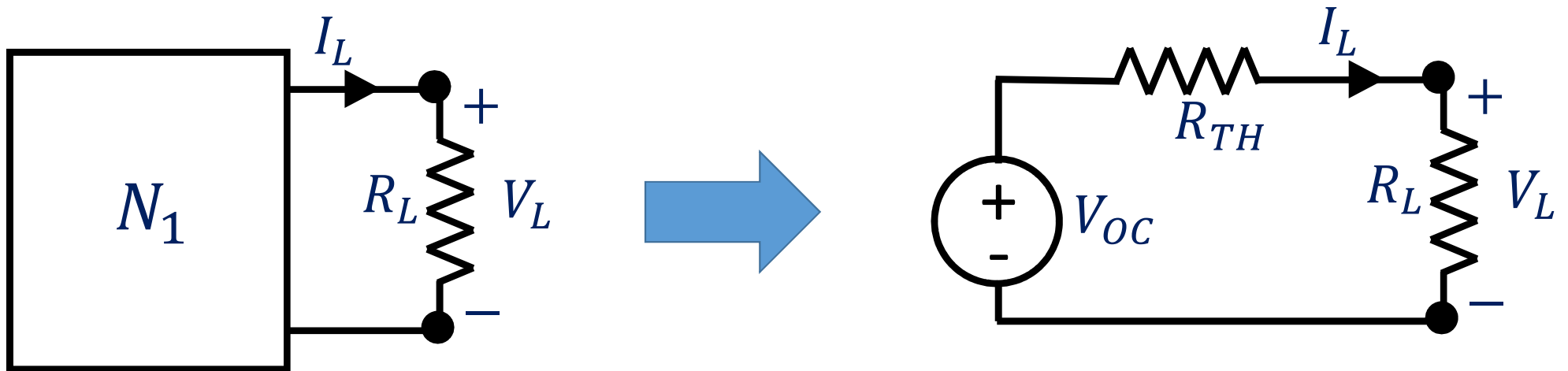
*Network Theorems: Thevenin's
and Norton's Theorem*

Tentative Lecture Plan

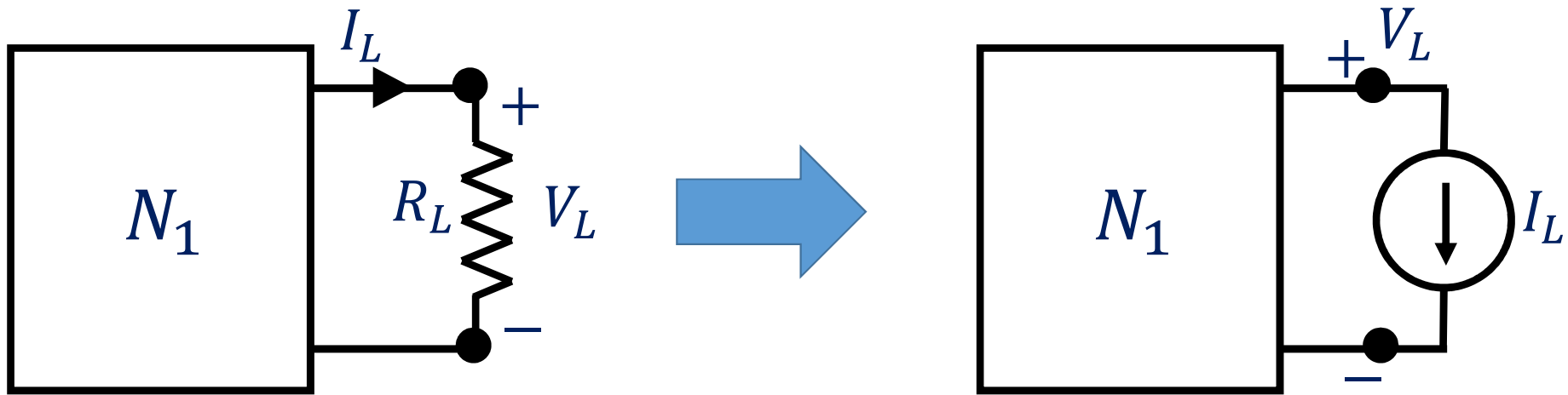
L	LDay-1	LDay-2	Topics	T	TDay
1	FRI, 04 AUG	MON, 07 AUG	Basic Definitions, Voltage & Current Sources, KCL & KVL	1	THU, 10 AUG
2	TUE, 08 AUG	FRI, 11 AUG	Nodal Analysis: Independent and Dependent Current Sources	2	SAT, 19 AUG
3	MON, 14 AUG	THU, 17 AUG	Nodal Analysis: Current and Voltage Sources		
4	FRI, 18 AUG	MON, 21 AUG	Mesh Analysis: Voltage and Current Sources	3	THU, 24 AUG
5	TUE, 22 AUG	FRI, 25 AUG	Network Theorems: Superposition Theorem	R	SAT, 26 AUG
SUN, 27 AUG : Quiz-1 : Lectures 1-5					
6	MON, 28 AUG	TUE, 29 AUG	Network Theorems: Thevenin's and Norton's Theorem	4	THU, 31 AUG
7	MON (Fr), 04 SEP	TUE, 05 SEP	Network Theorems: Maximum Power Transfer Theorem	5	WED, 06 SEP
8	FRI, 08 SEP	SAT (Mo), 09 SEP	Series RL and RC Circuits		
9	MON, 11 SEP	TUE, 12 SEP	Series RL, RC, RLC Circuits Response to Constant, Sinusoid and Exponential Input	6	THU, 14 SEP
10	FRI, 15 SEP	SAT (Tu), 16 SEP	Concept of Phasor, AC Circuits, AC Power		
WED, 20 SEP: MidSem : Lectures 1 – 10					

Thevenin's Theorem

Consider a Network N_1 containing all linear circuit elements and with an unique solution. Then, the interconnection between N_1 and a load resistance R_L may be replaced by an independent Voltage Source V_{OC} in series with a resistance R_{TH} without affecting any voltage across or current flowing through R_L .

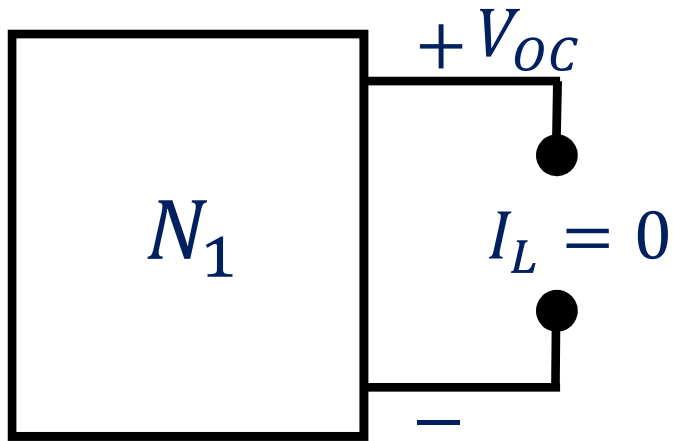


Thevenin's Theorem: Justification



V_L can be obtained by applying Superposition Theorem

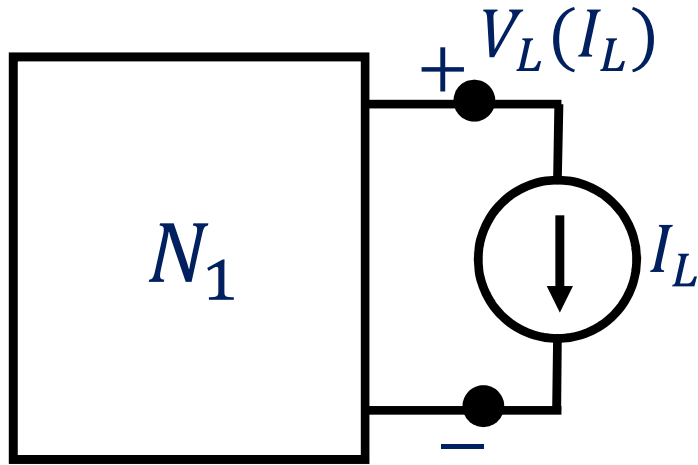
Thevenin's Theorem: Justification



Application of Superposition Theorem: Deactivate I_L and keep each Independent Source within N_1 active.

Superposition of Responses of all Independent Sources within N_1 provides $V_{OC} = V_L(I_L = 0)$

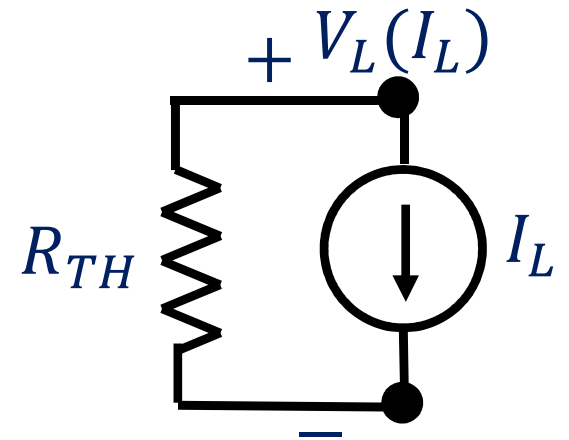
Thevenin's Theorem: Justification



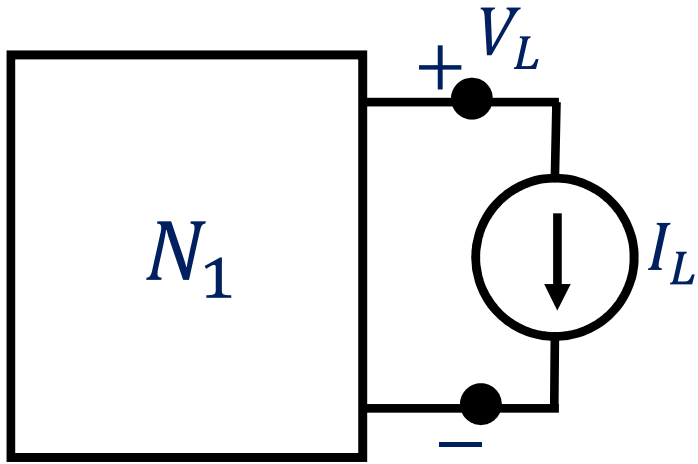
Application of KVL provides
 $V_L(I_L) + R_{TH}I_L = 0$

Application of Superposition Theorem:
Keep I_L active and Deactivate all
Independent Sources within N_1 .

N_1 is a Linear Network and is Sensed
as an Equivalent Resistance R_{TH}



Thevenin's Theorem: Justification

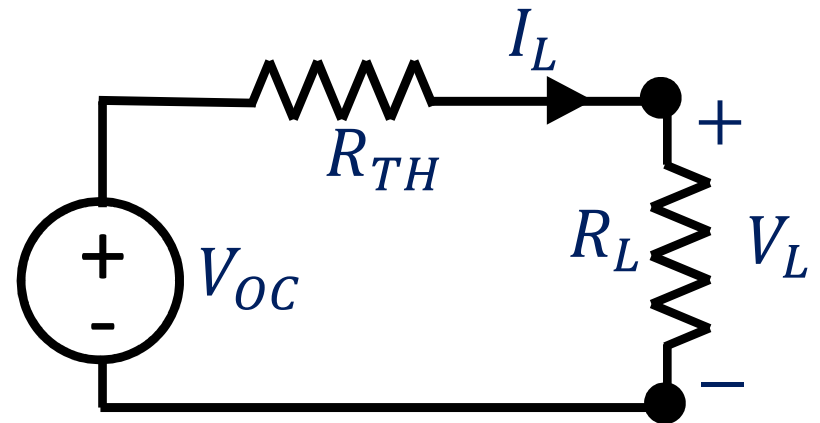


This can be accommodated if N_1 is replaced with independent voltage source V_{OC} in series with resistance R_{TH}

Application of Superposition Theorem:

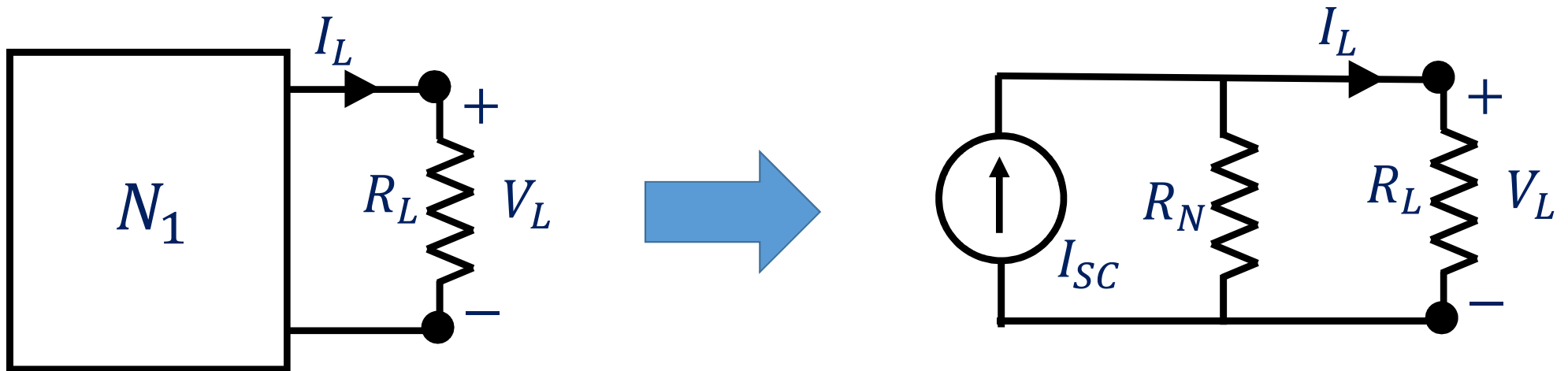
$$V_L = V_L(I_L = 0) + V_L(I_L) \text{ i.e.}$$

$$V_L = V_{OC} - R_{TH}I_L$$

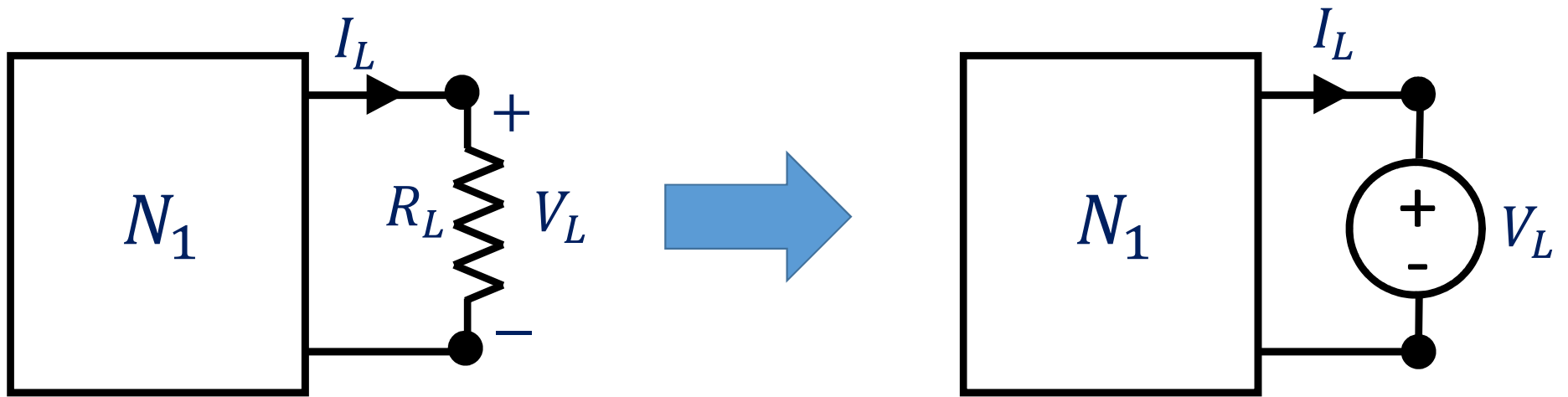


Norton's Theorem

Consider a Network N_1 containing all linear circuit elements and with a unique solution. Then, the interconnection between N_1 and a load resistance R_L may be replaced by an independent Current Source I_{SC} in parallel with a resistance R_N without affecting any voltage across or current flowing through R_L .

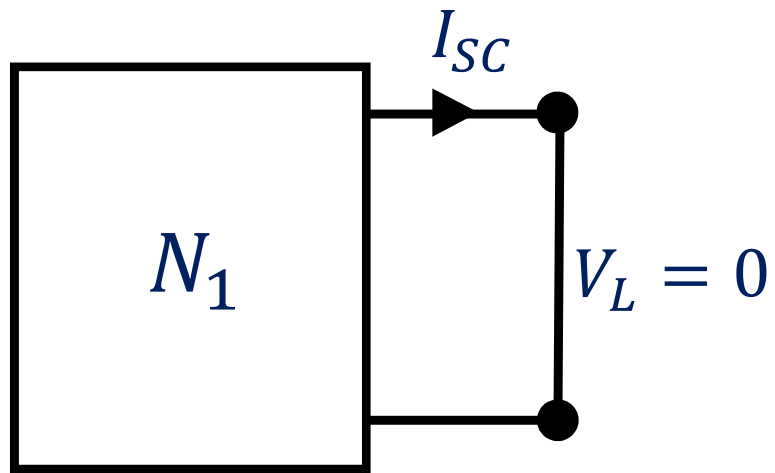


Norton's Theorem: Justification



I_L can be obtained by applying Superposition Theorem

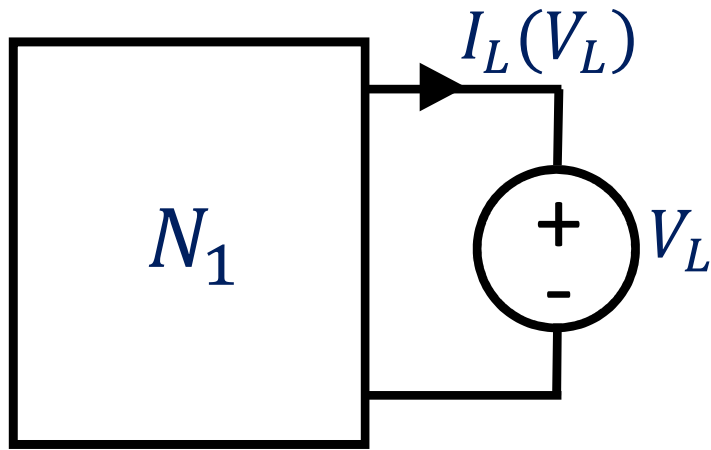
Norton's Theorem: Justification



Application of Superposition Theorem: Deactivate V_L and keep each Independent Source within N_1 active.

Superposition of Responses of all Independent Sources within N_1 provides $I_{SC} = I_L(V_L = 0)$

Norton's Theorem: Justification

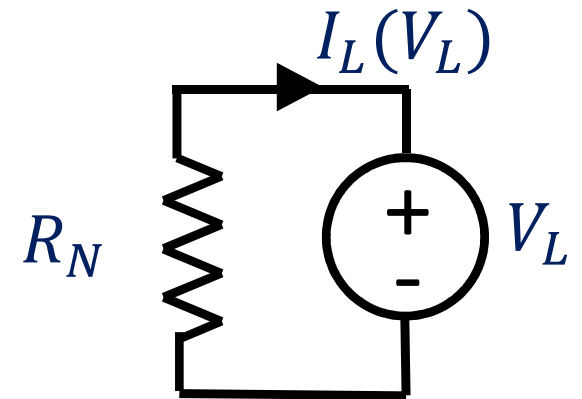


Application of KVL provides

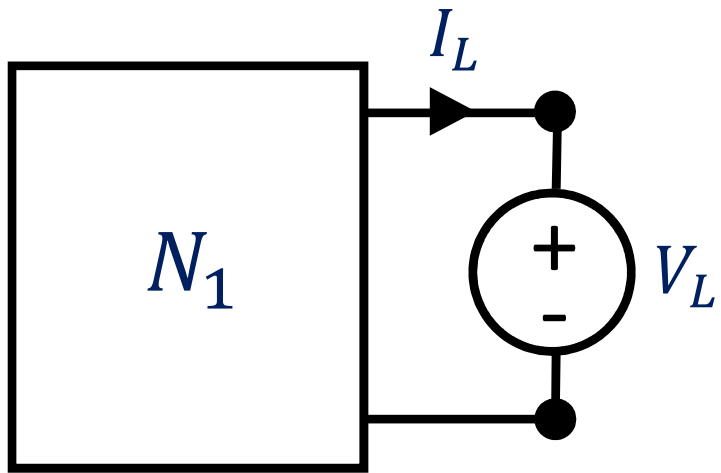
$$V_L + R_N I_L(V_L) = 0$$

Application of Superposition Theorem:
Keep V_L active and Deactivate all
Independent Sources within N_1 .

N_1 is a Linear Network and is Sensed
as an Equivalent Resistance R_N



Norton's Theorem: Justification

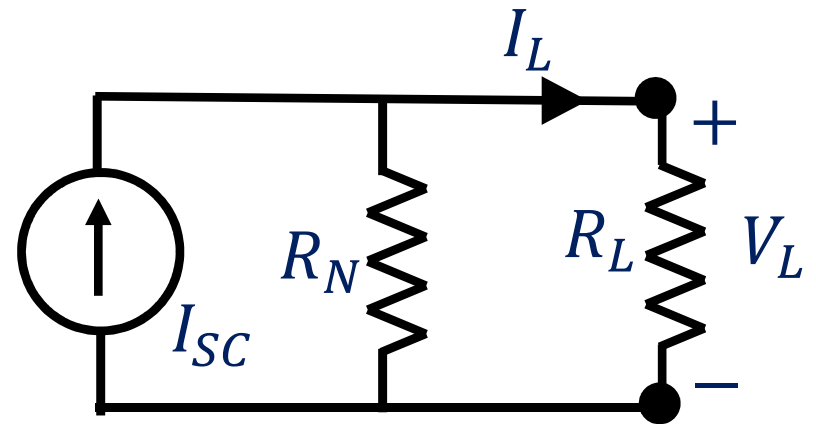


Application of Superposition Theorem:

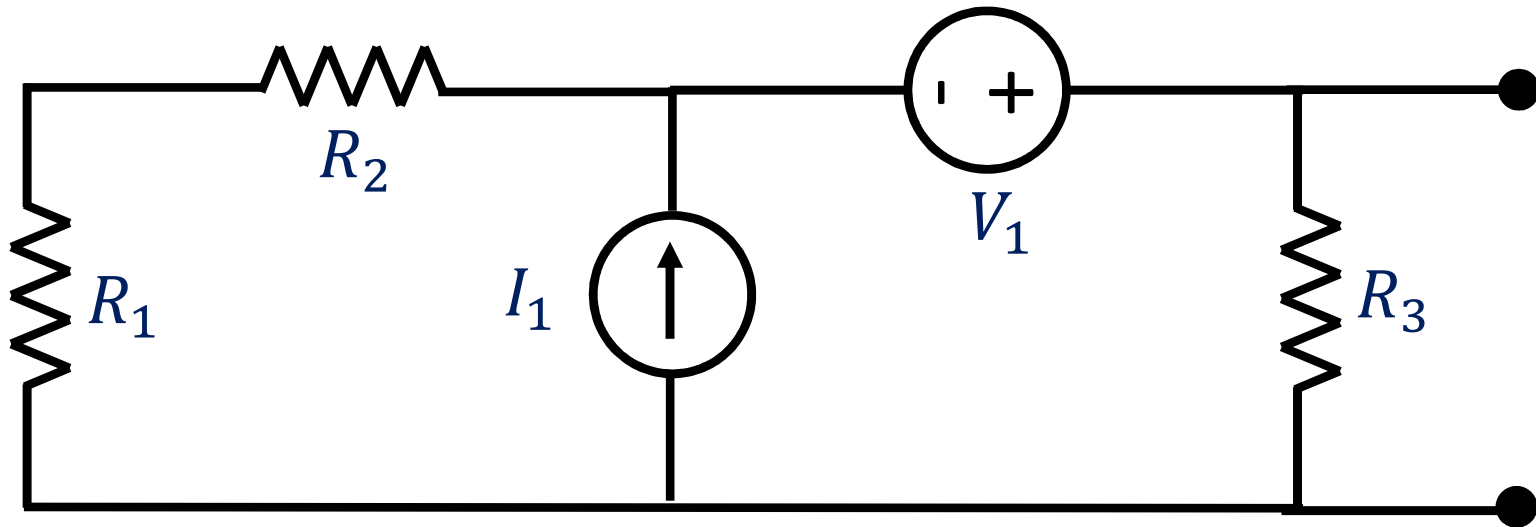
$$I_L = I_{SC}(V_L = 0) + I_L(V_L) \text{ i.e.}$$

$$I_L = I_{SC} - \frac{V_L}{R_N}$$

This can be accommodated if N_1 is replaced with independent current source I_{SC} in series with resistance R_N

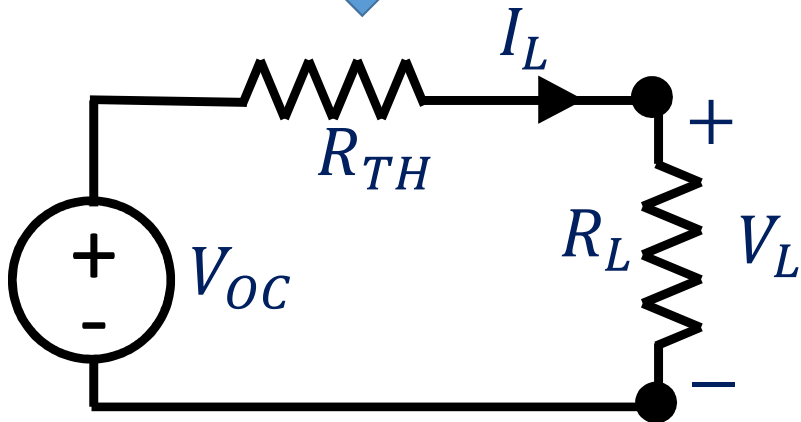
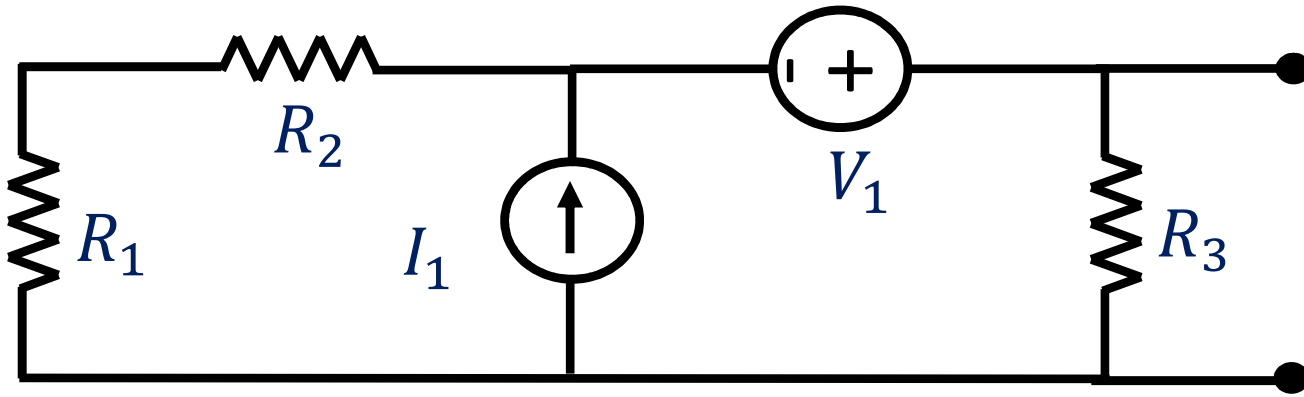


Draw Thevenin's/Norton's Equivalent Networks



$$R_1 = 1k\Omega \quad R_2 = 2k\Omega \quad I_1 = 2mA \quad V_1 = 3V \quad R_3 = 6k\Omega$$

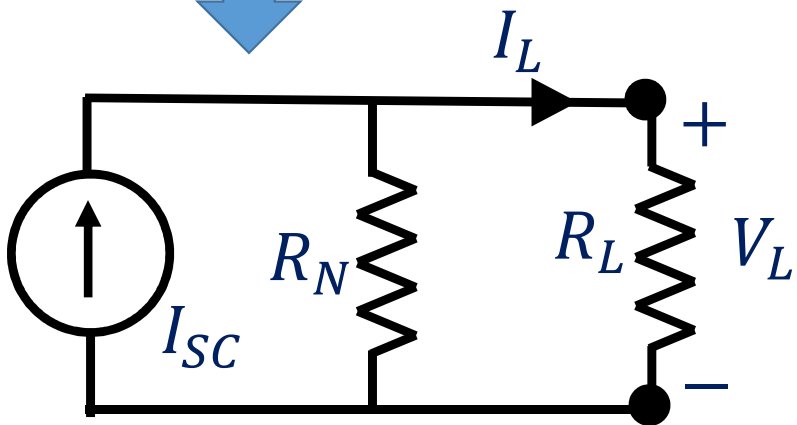
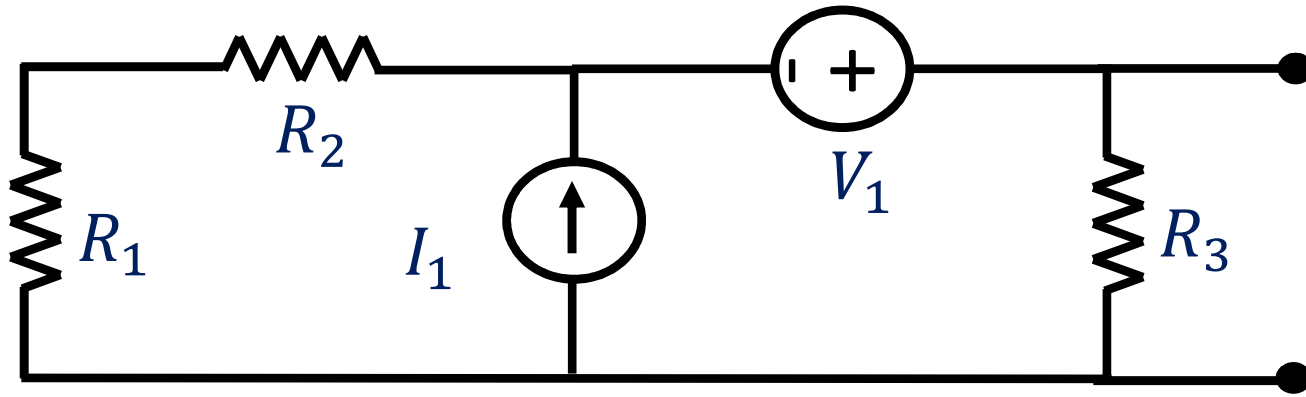
Thevenin's Equivalent Network



$$V_{OC} = \frac{R_3}{R_1 + R_2 + R_3} \{V_1 + (R_1 + R_2)I_1\} = 6V$$

$$R_{TH} = \frac{R_3(R_1 + R_2)}{R_1 + R_2 + R_3} = 2k\Omega$$

Norton's Equivalent Network



$$I_{SC} = \frac{V_1 + (R_1 + R_2)I_1}{R_1 + R_2} = 3mA$$

$$R_{TH} = \frac{R_3(R_1 + R_2)}{R_1 + R_2 + R_3} = 2k\Omega$$

Evaluate V_{OC} , I_{SC} and R_{TH}/R_N

$$R_1 = 0.2 \, \Omega$$

$$R_2 = 1.0 \, \Omega$$

$$R_3 = 0.5 \, \Omega$$

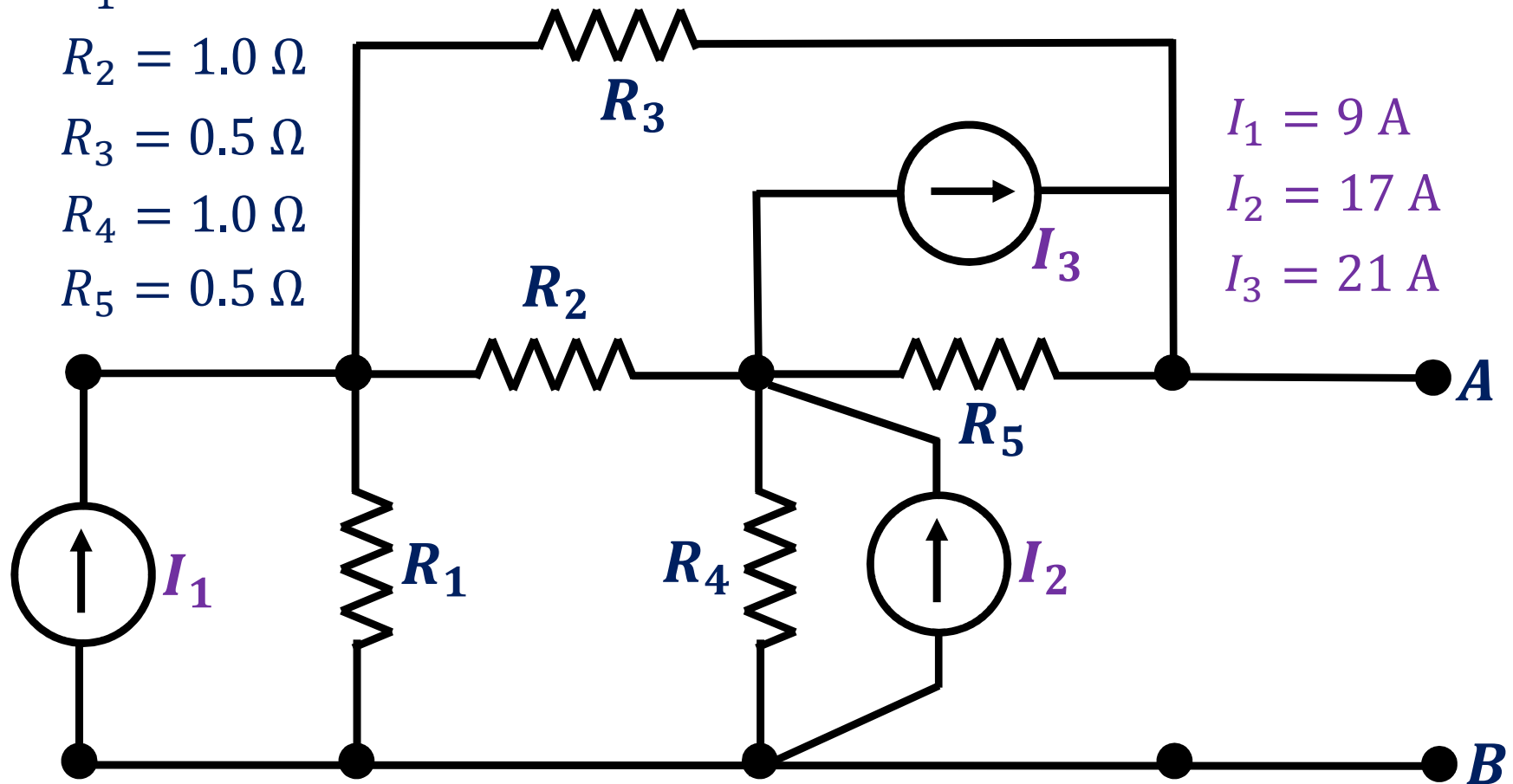
$$R_4 = 1.0 \, \Omega$$

$$R_5 = 0.5 \, \Omega$$

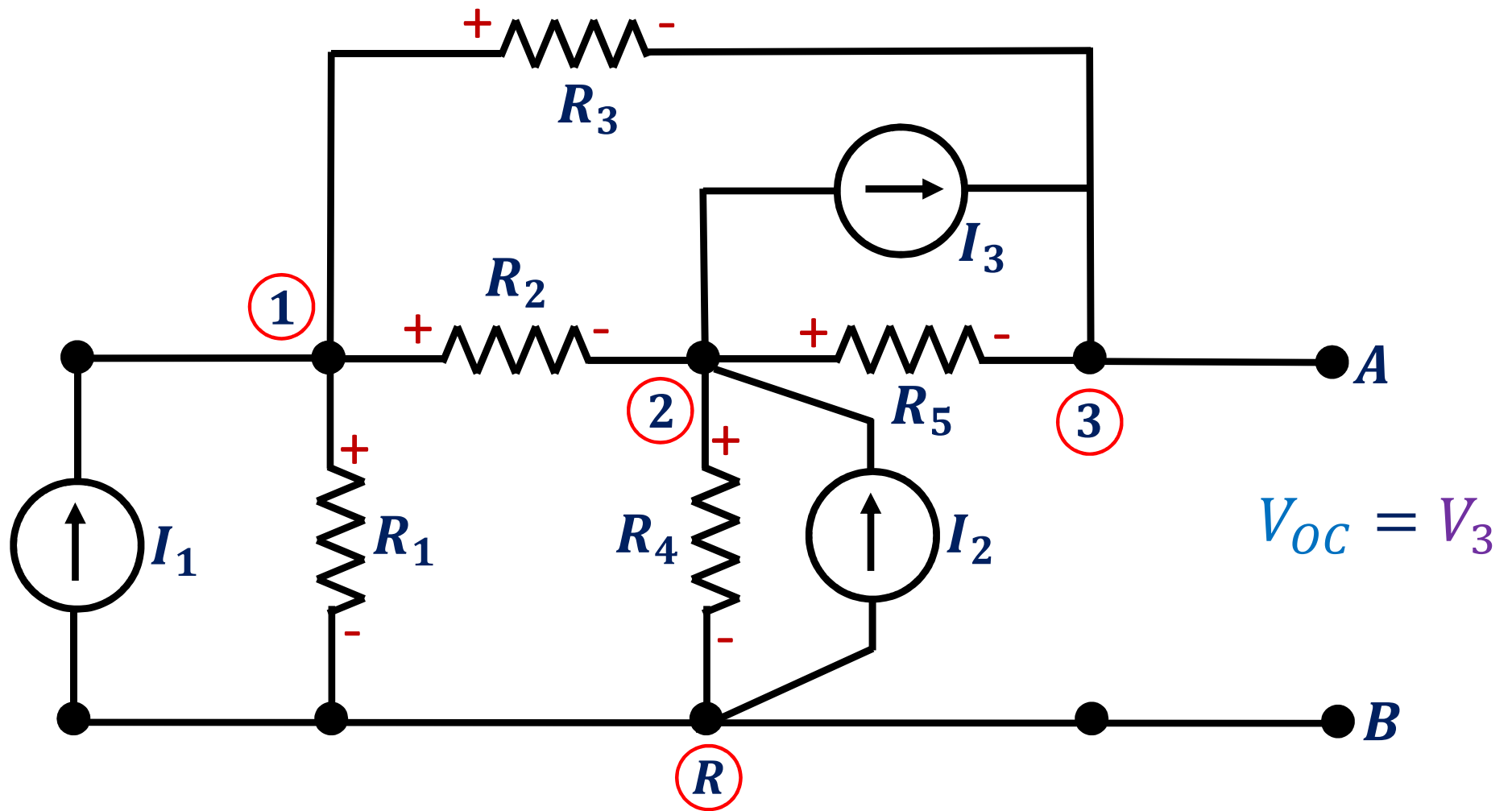
$$I_1 = 9 \, \text{A}$$

$$I_2 = 17 \, \text{A}$$

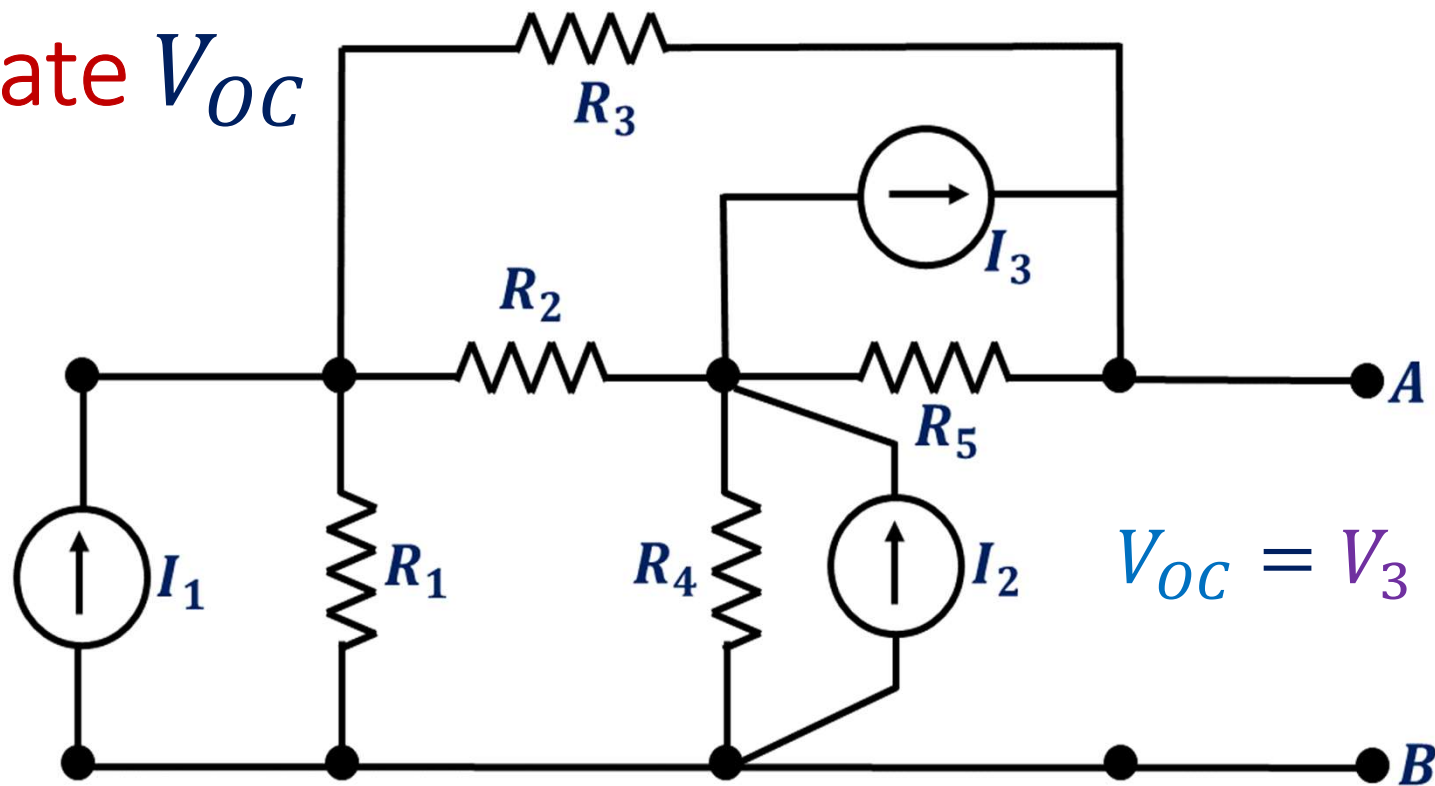
$$I_3 = 21 \, \text{A}$$



Evaluate V_{OC}



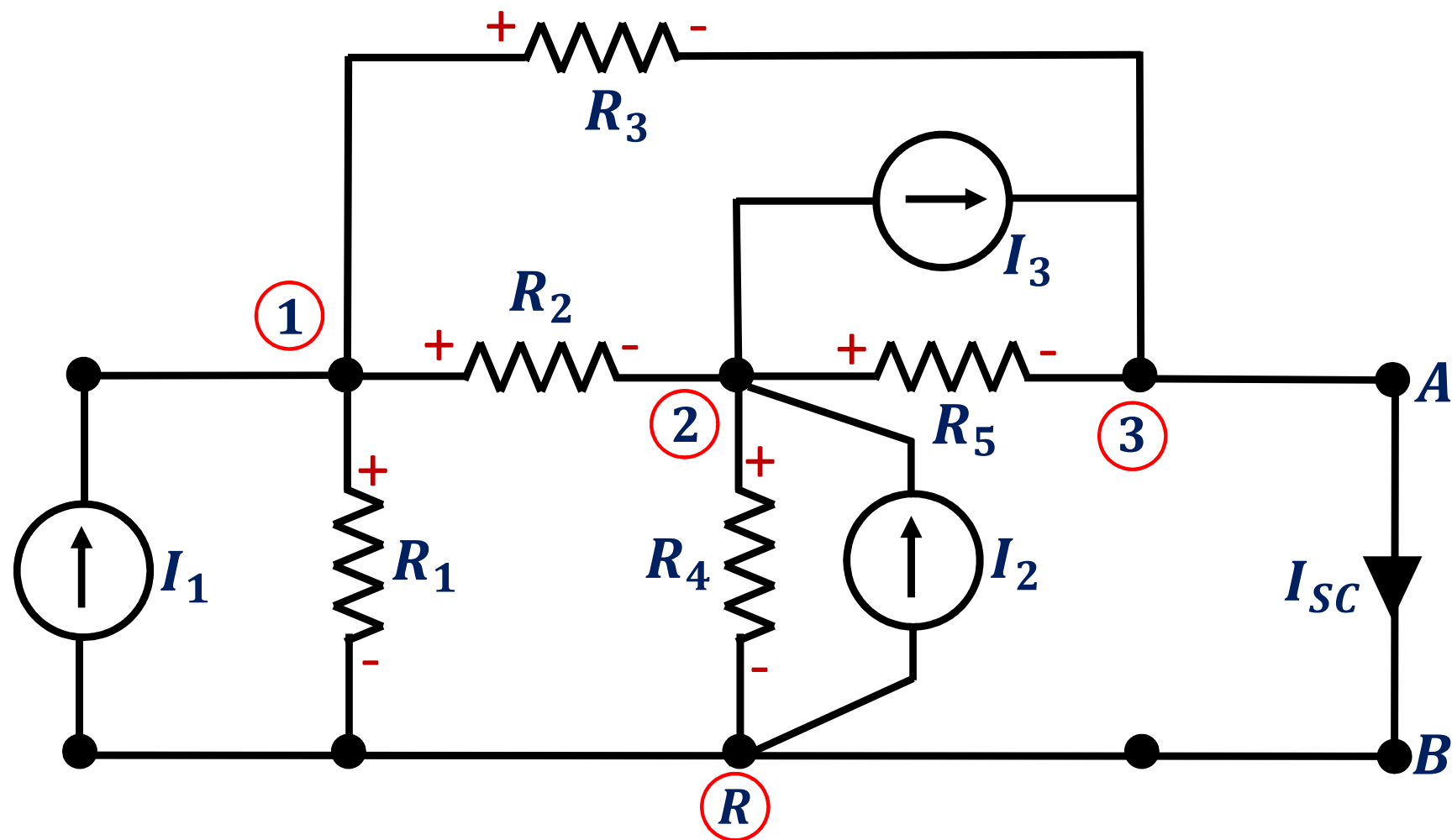
Evaluate V_{OC}



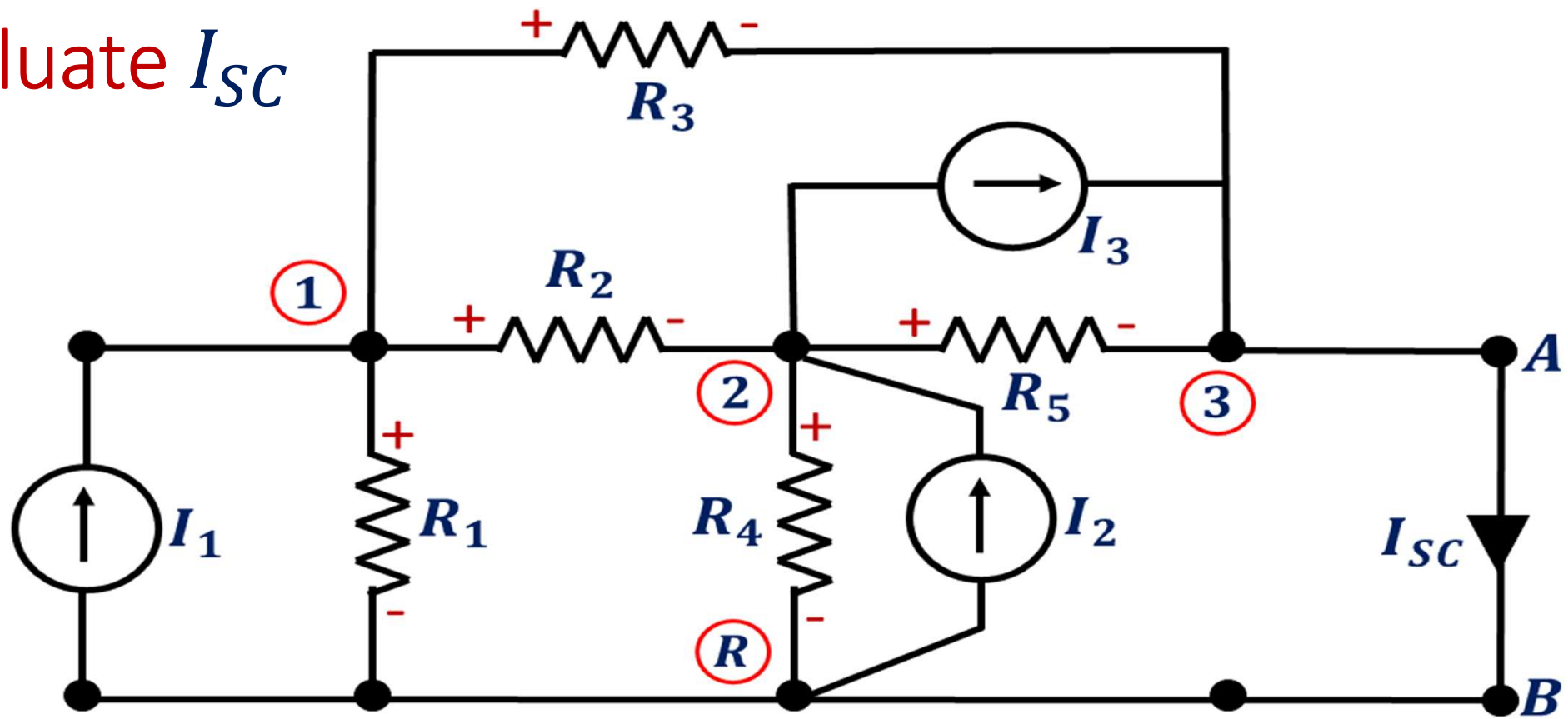
$$V_{OC} = V_3 = 9.838V$$

$$\begin{bmatrix} G_1 + G_2 + G_3 & -G_2 & -G_3 \\ -G_2 & G_2 + G_4 + G_5 & -G_5 \\ -G_3 & -G_5 & G_3 + G_5 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix}$$

Evaluate I_{sc}

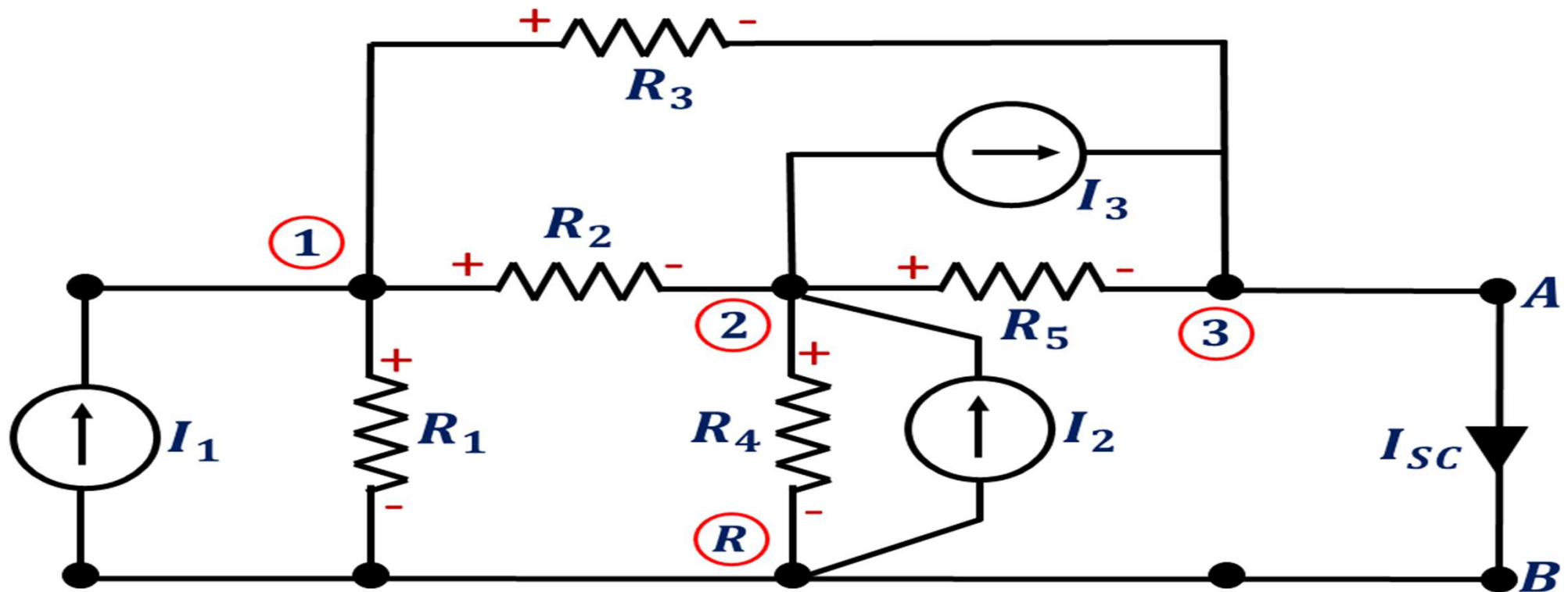


Evaluate I_{SC}



$$\begin{bmatrix} G_1 + G_2 + G_3 & -G_2 \\ -G_2 & G_2 + G_4 + G_5 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix}$$

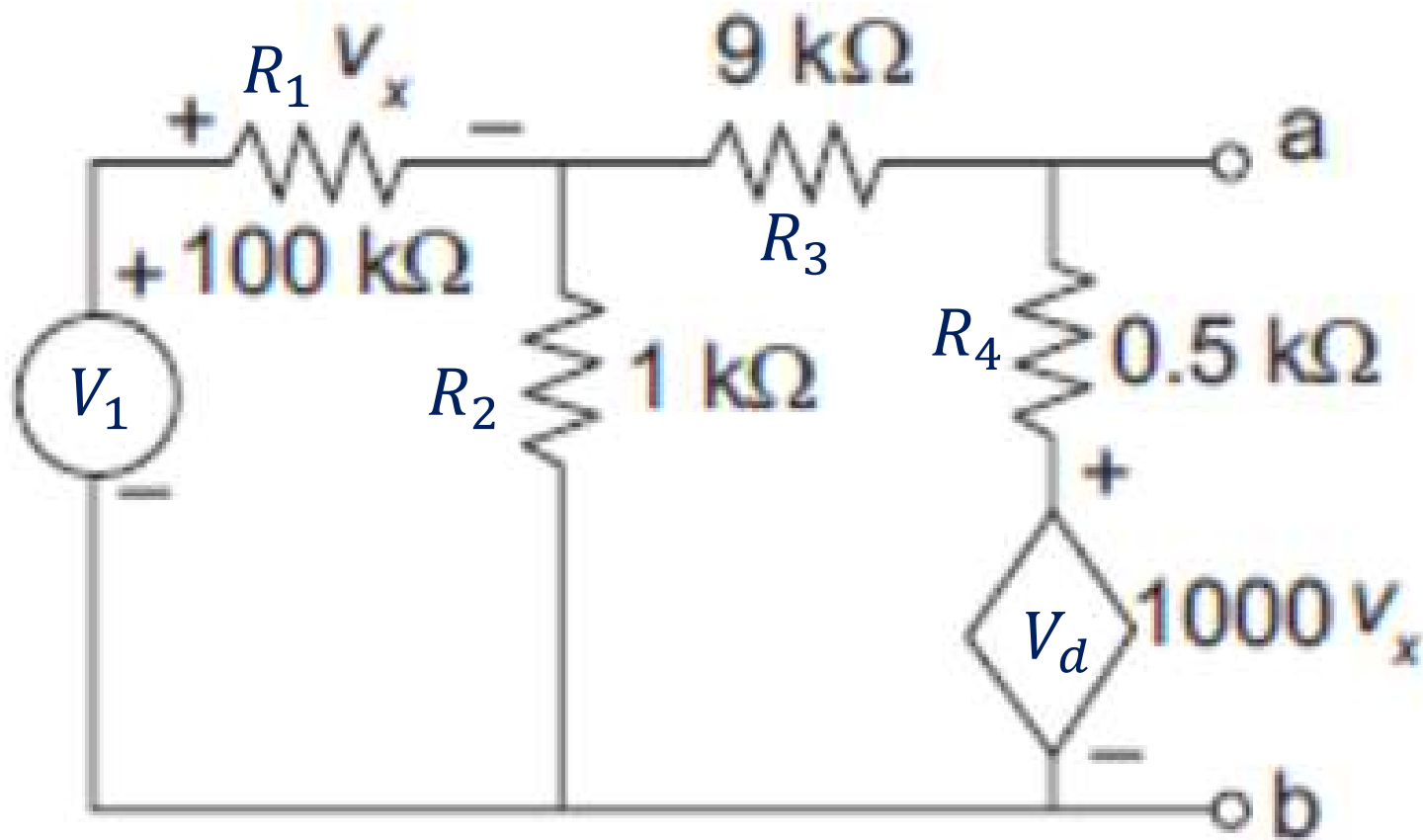
Evaluate I_{SC} and R_{TH}/R_N



$$I_{SC} = I_3 + G_3 V_1 + G_5 V_2 = 21.580 \text{ A}$$

$$R_{TH} = \frac{V_{OC}}{I_{SC}} = 0.456 \Omega$$

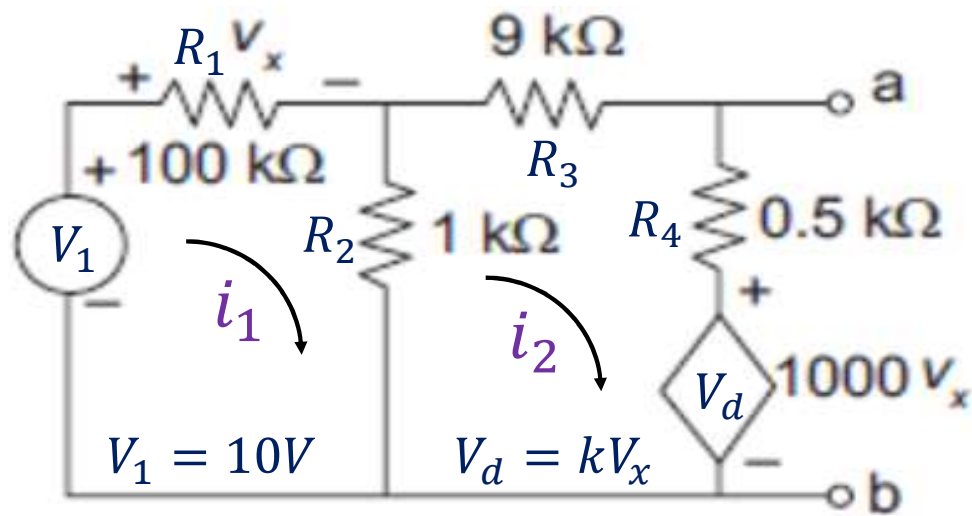
Draw Thevenin's Equivalent Network



$$V_1 = 10V$$

$$V_d = kV_x$$

Draw Thevenin's Equivalent Network



$$i_1 = 0.000001\text{A}$$

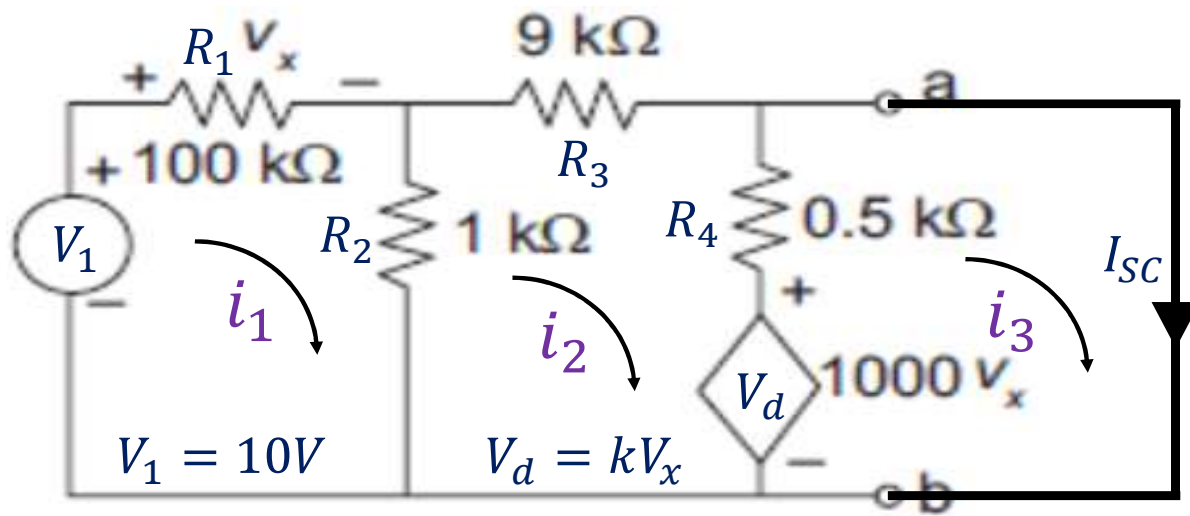
$$i_2 = -0.009895\text{A}$$

$$V_{OC} = 98.953\text{V}$$

$$\begin{bmatrix} R_1 + R_2 & -R_2 \\ kR_1 - R_2 & R_2 + R_3 + R_4 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} V_1 \\ 0 \end{bmatrix}$$

$$V_{OC} = R_4 i_2 + kR_1 i_1$$

Draw Thevenin's Equivalent Network



$$R_{TH} = \frac{V_{OC}}{I_{SC}}$$

$$R_{TH} = 4.992\Omega$$

$$\begin{bmatrix} R_1 + R_2 & -R_2 & 0 \\ kR_1 - R_2 & R_2 + R_3 + R_4 & -R_4 \\ -kR_1 & -R_4 & R_4 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} [V_1]$$

$$I_{SC} = i_3 = 19.822A$$

Summary

- Thevenin's Theorem
- Norton's Theorem
- Circuits with
 - Independent Sources
 - Dependent Sources



Thank You